Event-Gated Ensemble Enhancement with ASR-Consistent Training for Indic Noise Event Removal

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Abstract—We present our Track 2 system for the Datathon@IndoML 2026 noise event removal challenge: suppressing non-speech events (horns, animals, breathing, children, alarms, machinery, music) in real-world Indic speech from the Vaani corpus, scored by SI-SDR on synthetic clips plus 100× the WER reduction of the frozen mandated ASR (SraVaani-1.0) over all clips. A single learned enhancer is applied identically to every clip. It averages a waveform-domain DNS64 (Demucs) model and a band-split RNN with a learned frame-level event gate, both trained on on-the-fly mixtures whose event statistics follow the validation set and with an ASR-consistency loss through the frozen SraVaani encoder. A second stage fine-tunes both models on longer, denser and louder mixtures, with checkpoints selected on a pseudo-test built from held-out validation clips that predicted our leaderboard score within 0.13. Our final submission (Codabench 995085) reaches SI-SDR 12.78 dB and ΔWER +1.87 (synthetic +5.41), a combined score of 14.65 (Phase 2 rank 13), with the hidden ASR audit passed.

Index Terms—speech enhancement, noise event removal, band-split RNN, Demucs, ASR-aware training, Indic speech

I. Training Data and Preprocessing

A. Data sources

Table I lists every source. From the Vaani Noise Event Dataset [1] (90,637 segments, 154.6 h, Gold/Silver/Bronze tiers) we keep only clean speech: non-event regions of at least 1.5 s, at least 0.25 s from any labelled event, from clips whose 10th-percentile frame energy lies more than 28 dB below the clip RMS. Of the IndoML validation set (2,501 clips; 1,806 natural, 695 synthetic with clean references), the 695 synthetic pairs are split 560/135 into training and holdout. Noise events are cut from noisy − clean at the labelled spans (±50 ms) of the 560 training pairs; in these pairs the added noise lies entirely inside the spans. The external ESC-50 [6] clips (resampled to 16 kHz, trimmed to the region within 30 dB of the peak) make the model generalise beyond the validation noise, and Indic FLEURS [7] shards add clean read speech. All banks are stored as memory-mapped float16 arrays.

TABLE I
Training data sources (external data marked †).

Source	Amount used	Role
Vaani Noise Event (Gold+Silver)	8 h	clean speech (low-background regions)
IndoML validation, 560 synth. pairs	0.81 h clean, 1,783 events	clean speech; noise events
IndoML validation, 135 holdout	106-clip pseudo-test	checkpoint and submission selection only
ESC-50† [6]	2,000 clips	noise events (p=0.3)
FLEURS† [7], 12 Indic languages	≤600 utt./lang., ≤12 s	clean read speech
Totals: 3,783 noise events (2.72 h); up to 17.6 h of clean speech.		

B. Augmentation pipeline

Mixtures are generated on the fly, so every batch is new:

1) Speech. A validation reference (p=0.5) or a Vaani/FLEURS segment. Utterances rescaled within ±3 dB and separated by 50–600 ms gaps are joined until the 6–10 s crop is full; the crop is scaled to $-30 \ldots -15$ dBFS.
2) Event count $k = n \cdot U(1, L/4.5\,\text{s}) \cdot U(\text{ev_mult})$, with n drawn from the per-clip counts of the validation synthetic set.
3) Events. A validation noise event (p=0.7) or an ESC-50 crop of empirical duration (p=0.3), with 5 ms fades and random polarity at a uniform offset. Its level relative to the speech RMS follows the empirical validation distribution (median +4.3 dB) $+\mathcal{N}(0, 2)$ dB $+U(\text{db_shift})$.
4) Targets. Clean speech and the frame event mask (hop 128); mixture and target are peak-normalised jointly.

The simulated mixtures match validation (median input SI-SDR −2.5 vs. −2.6 dB; event coverage 43% vs. 45%). Stage 2 uses ev_mult=$(1, 2)$, db_shift=$(0, 3)$ dB and 10 s crops.

II. Proposed Method

A. Architecture

Fig. 1 shows the system.

DNS64. The Demucs waveform U-Net with an LSTM bottleneck [2] (hidden width 64, 33.5 M parameters), initialised from the public checkpoint (Table II).

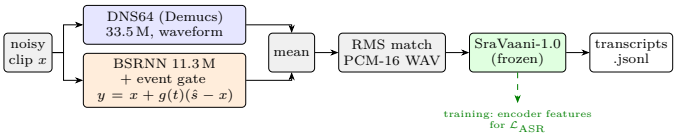


Fig. 1. Inference: every clip passes through both enhancers, whose outputs are averaged and loudness-matched to the input; the mandated ASR then transcribes the written WAVs. During training its frozen encoder also provides $\mathcal{L}_{\text{ASR}}$.

TABLE II
Pretrained models and their roles.

Model	Version and source	Role
DNS64 [2]	dns64-a7761ff99a7d5bb6.th, facebookresearch/denoiser (CC BY-NC 4.0)	initialisation of the DNS64 member
SraVaani-1.0 [8]	ARTPARK-IISc/SraVaani-1.0, Hugging Face, unmodified	frozen encoder for $\mathcal{L}_{\text{ASR}}$; transcription

Event-gated BSRNN (ours, 11.3 M parameters), after band-split RNN [3]. A 512-point STFT (hop 128) is split into 37 bands (16×4, 16×8, 4×16 and 1 bins), each normalised and projected to 112 features. Six blocks alternate residual BLSTMs over time and over bands, and per-band GLU MLPs output a complex mask $\mathbf{M}$, giving $\hat{s} = \text{iSTFT}(\mathbf{M} \odot \mathbf{X})$. A gate head maps the band features to an event probability $p(t)$ and the output is

$$y = x + g(t)\,(\hat{s} - x), \qquad g(t) = \gamma + (1 - \gamma)\,p(t), \tag{1}$$

with floor $\gamma = 0.2$, so every frame is processed while event-free speech stays close to the input.

B. Losses

For mixture x, clean target s and event mask m,

$$\mathcal{L}_{\text{BSRNN}} = \tfrac{1}{2}\mathcal{L}_{\text{rec}}(\hat{s}, s) + \tfrac{1}{2}\mathcal{L}_{\text{rec}}(y, s) + \text{BCE}(p, m) + \lambda\,\mathcal{L}_{\text{ASR}}, \tag{2}$$

$$\mathcal{L}_{\text{rec}}(a, s) = -\,\text{SI-SDR}(a, s) + 0.5\,\text{MRSTFT}(a, s), \tag{3}$$

and $\mathcal{L}_{\text{DNS64}} = \mathcal{L}_{\text{rec}}(\hat{s}, s) + \lambda\mathcal{L}_{\text{ASR}}$. SI-SDR is the metric itself [4]; MRSTFT is the L1 distance between log-magnitudes at FFT sizes 256, 512 and 1024 [5]. The ASR-consistency loss $\mathcal{L}_{\text{ASR}}$ is the L1 distance between the frozen SraVaani encoder outputs on the enhanced and clean signals, normalised by the target's mean magnitude, applied every second step with $\lambda = 0.1$. Its front end replicates the ASR's pre-emphasis, 128-bin log-mel and per-feature normalisation, and gradients flow through the frozen encoder, so the enhancer learns outputs that the mandated ASR transcribes like clean speech.

C. Training strategy

All runs use AdamW [10] (weight decay 10^{-2}), a 500-step warm-up and a cosine schedule driven by wall-clock time, which guarantees completion inside a fixed GPU allocation, with gradient clipping at 5 and bf16 autocast for BSRNN. Table III lists the runs.

TABLE III
Training runs (batch 16 unless noted). Stage 1 is selected on the 135-clip holdout, stage 2 on the pseudo-test.

Run	Init	Crop	LR	λ	Min.	GPU
DNS64 s1	DNS64	6 s	10^{-4}	0.1	210	B200
BSRNN s1	scratch	6 s	10^{-3}	0.1	180	A100 80 GB
DNS64-b s1	DNS64	8 s	$2{\cdot}10^{-4}$	0.2	150	A100 80 GB
DNS64 s2	DNS64 s1	10 s	$5{\cdot}10^{-5}$	0.1	50	B200
BSRNN s2	BSRNN s1	10 s	$2{\cdot}10^{-4}$	0.1	50	B200 (batch 8)

TABLE IV
SI-SDR (dB) on the validation holdout (135 clips) and the pseudo-test (106 clips), and the Phase 2 test score (SI-SDR / ΔWER / combined).

System	Holdout	Pseudo-test	Test (Codabench)
Noisy input	−0.76	−3.75	–
DNS64 (stage 1)	19.04	12.81	–
BSRNN, gated (stage 1)	19.04	12.67	–
DNS64 + BSRNN	19.78	13.36	12.66 / 1.80 / 14.46
+ second DNS64	19.71	13.36	– / – / 14.41
Stage-2 DNS64 + BSRNN	–	13.52	12.78 / 1.87 / 14.65

Pseudo-test. After our first submission revealed a large gap between validation and test, we built a pseudo-test only from the 135 holdout clips: 106 clips of 2–3 joined holdout references (median 13.5 s) carrying their own noise plus 0.5× extra holdout events at +1.5 dB. Its median input SI-SDR (−4.5 dB) matches the level we estimated for the real synthetic test clips (−4.48 dB), and it reproduces the per-clip score distribution of our first submission (median 12.79 vs. 12.27 dB; 5th percentile 4.78 vs. 4.59 dB). Stage 2 fine-tunes DNS64 and BSRNN on these test-like conditions (BSRNN's clean pool excludes Vaani segments).

Inference. The final system averages the two stage-2 models (gate floor 0.2), matches the output RMS to the input, processes 20 s chunks with a 2 s cross-fade and writes PCM-16. Transcripts come from the unmodified SraVaani-1.0 (fp32, greedy TDT [9]) on the written WAVs.

Candidate selection. Candidates are ranked by an estimated leaderboard score: pseudo-test SI-SDR minus a 0.70 dB calibration offset, plus 0.32× the synthetic ΔWER (holdout clips, against ASR-on-clean transcripts, calibrated to our first submission), plus 0.68× the natural-validation ΔWER; 0.32 and 0.68 are the synthetic and natural shares of words in the test set.

III. Results and Conclusion

Table IV and Figs. 2–3 summarise the results.

- Final submission (995085): SI-SDR 12.78 dB and ΔWER +1.87 (+5.41 on synthetic, +0.21 on natural clips), combined 14.65; the hidden ASR audit passed with 0% mismatch.

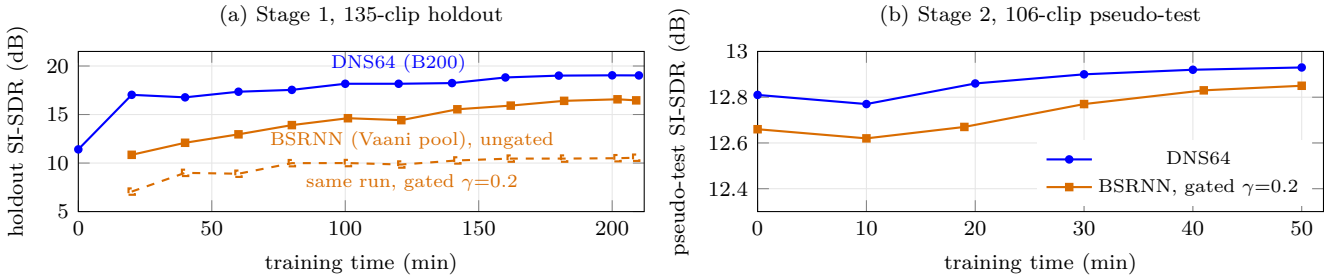


Fig. 2. Evaluation SI-SDR during training, from the training logs. (a) Stage-1 DNS64 starts at 11.4 dB (the public DNS64 checkpoint) and reaches 19.0 dB. The BSRNN variant shown was trained with a clean pool that also contained noisier Vaani speech; with that pool the gate hurts (10.5 vs. 16.5 dB), unlike the final BSRNN trained on cleaner speech (Table IV). (b) Stage-2 fine-tuning on test-like mixtures; both models improve steadily on the pseudo-test.

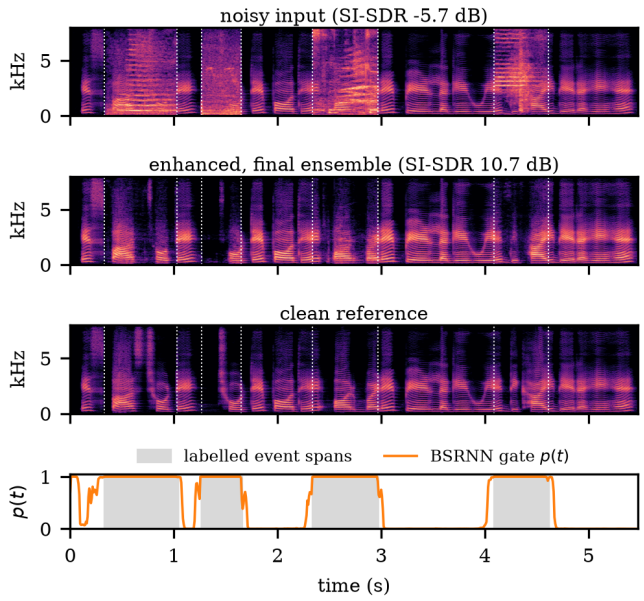


Fig. 3. The final system on a held-out validation clip (5.5 s; horn, bird squawking, child yelling, horn). Dotted lines mark the labelled event spans. The ensemble raises SI-SDR from −5.7 to 10.7 dB, and the BSRNN gate $p(t)$ (bottom) switches on over each event and stays near zero on event-free speech.

- Architectural diversity helps. DNS64 + BSRNN gains 0.75 dB over either model alone and turns the hold-out ΔWER positive, whereas averaging two DNS64 models adds nothing.
- The learned gate needs clean training speech. The final BSRNN gated at $\gamma = 0.2$ reaches 19.04 dB against 17.82 dB ungated; with a noisier clean pool the gate lowers SI-SDR instead (Fig. 2a), consistent with activity targets that mislabel real background noise as speech.
- Noise level, not duration, limits performance. On the scored test clips, per-clip SI-SDR correlates 0.82 with the estimated input SNR and 0.09 with clip length. Clips below −6 dB input SNR (39%) average 8.5 dB, and clips above 0 dB average 20.7 dB. On the pseudo-test our ensemble (13.36 dB) already exceeds the ideal ratio mask (12.99 dB) and approaches the ideal phase-sensitive mask (15.15 dB), so mask-style enhancement has little headroom left at such SNRs.
- Pseudo-test fidelity. The estimator predicted 14.52 for our final entry, which scored 14.65, whereas validation-holdout estimates were optimistic by about 5 dB because test clips are 2.7× longer and much noisier.

Novelty. (i) An event-gated BSRNN that learns event activity jointly with separation; (ii) ASR-consistency training through the frozen mandated ASR; (iii) wall-clock-scheduled training with validation-matched on-the-fly mixing; and (iv) a holdout-derived pseudo-test with a calibrated estimate of the leaderboard score, used to choose submissions.

Conclusion. A single learned, event-aware enhancer trained on statistically matched mixtures gives strong SI-SDR and a positive ASR gain on natural and synthetic speech; very low-SNR clips remain the main gap.

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